

1 Summary

Model (Run-name): model1 (test)

Iteration: 8

TAC: 153386.34908473463

Took: 1.8953087329864502s

Total Time: 7.156022310256958s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

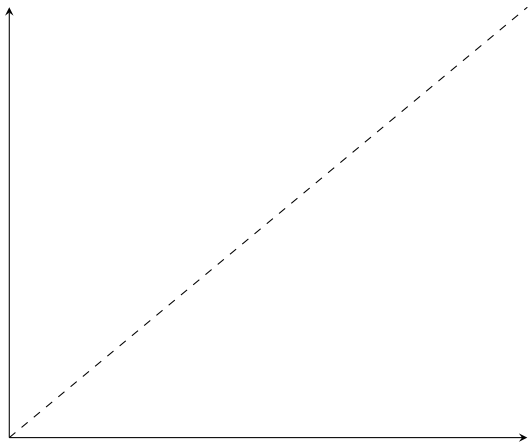
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

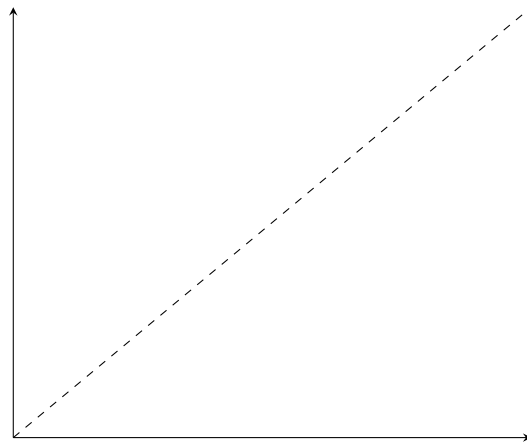
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05202106736660487	$b_{1,1,1}^{H,out}$	906.9632139432064
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017780776513256547	$b_{1,2,2}^{H,out}$	3863.64945971849
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.028096007654785208	$b_{2,1,2}^{H,out}$	9379.525810422025
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.0146495744298314	$b_{1,1,1}^{C,in}$	8303.29988183539
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009563349339146031	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.02250701784425265	$b_{2,1,2}^{C,in}$	4611.408893509331
$A_{1,1,1}^\beta$	70.30181758862284	$q_{1,1,1}$	686.3505402813454	$b_{1,1,1}^{C,out}$	8989.650422116736
$A_{1,2,2}^\beta$	69.34502840170047	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	131.69760457629607	$q_{2,1,2}$	2420.4741895779807	$b_{2,1,2}^{C,out}$	7031.883083087312
$A_{cu,1}^\beta$	3.1809419553840232	$q_{cu,1}$	163.6494597186546		
$A_{cu,2}^\beta$	37.47875108132638	$q_{cu,2}$	1979.5258104220193		
$A_{hu,1}^\beta$	13.22717920580809	$q_{hu,1}$	493.1752701406731		
$A_{1,1,1}$	70.30181758862284	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.34502840170047	$t_{1,2}^{(H)}$	581.3649459718655		
$A_{2,1,2}$	131.69760457629607	$t_{1,3}^{(H)}$	386.36494597186544		
$A_{cu,1}$	3.1809419553840232	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.47875108132638	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.22717920580809	$t_{2,3}^{(H)}$	468.97629052110096		
$dt_{1,1,1}$	32.87835134271154	$t_{1,1}^{(C)}$	617.1216486572885		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.3649459718654		
$dt_{1,2,2}$	81.3649459718655	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.36494597186547	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.63505402813462	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.976290521100964	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.36494597186547	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.97629052110096	$t_{1,2,2}^H$	386.3649459718548		
$dt_{hu,1}$	62.87835134271154	$t_{2,1,2}^H$	468.9762905211007		
$f_{1,1,1}^H$	2.4512519295762334	$t_{1,1,1}^C$	618.6342496050581		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	613.8951066390871		
$f_{1,1,1}^C$	14.531446372159014	$b_{1,1,1}^{H,in}$	1593.3137542245518		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5813.64945971849		
$f_{2,1,2}^C$	11.247338764656906	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

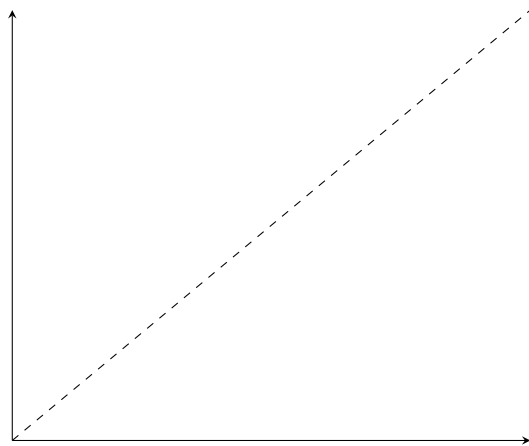
- $(1, 1, 1)$
 - Value: 0.052021
 - Error: 0.000003194192389
 - Relative error: 0.000061398130273
- $(1, 2, 2)$
 - Value: 0.017781
 - Error: 0.000115651335702
 - Relative error: 0.006462258092957
- $(2, 1, 2)$
 - Value: 0.028096
 - Error: 0.000462637927549
 - Relative error: 0.016199575229002
- $cu, 1$
 - Value: 0.014650
 - Error: 0.000020426803888
 - Relative error: 0.001392420052524
- $cu, 2$
 - Value: 0.009563
 - Error: 0.000000178173118
 - Relative error: 0.000018630480998
- $hu, 1$
 - Value: 0.022507
 - Error: 0.000000331727418
 - Relative error: 0.000014738626453



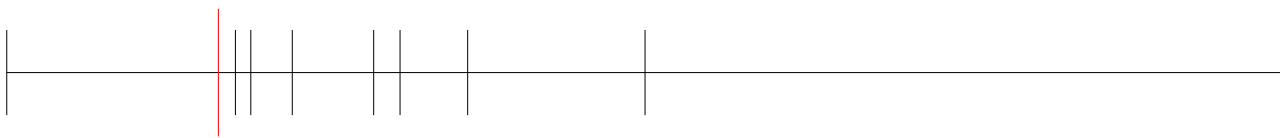
(a) *RecLMTD* - (1, 1, 1)



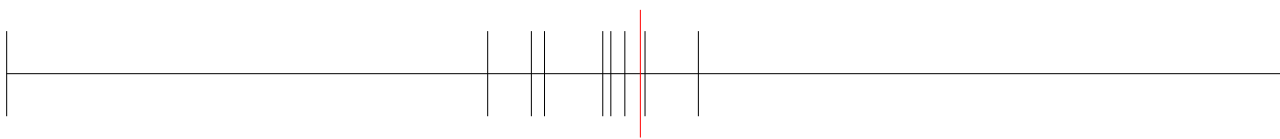
(b) *RecLMTD* - (1, 2, 2)



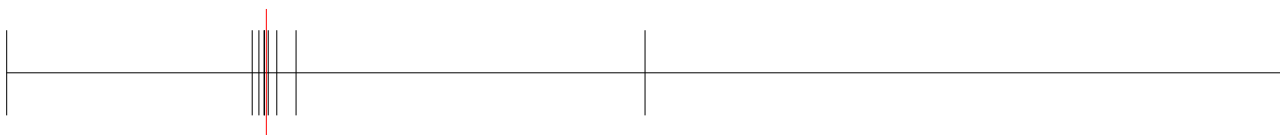
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1593.3137542245518
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5813.64945971849
 - Absolute error: 1.646185410209e-10
 - Relative error: -2.8315869775345303e-14
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 906.9632139432064
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3863.64945971849
 - Absolute error: 5.820766091346741e-11
 - Relative error: -1.5065461170928174e-14
- (2, 1, 2)
 - Value: 9379.525810422025
 - Absolute error: 1.0913936421275139e-11
 - Relative error: 1.163591490856411e-15

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8303.29988183539
 - Absolute error: 0.5408105136957602
 - Relative error: 6.513624074240058e-05
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4611.408893509331
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 8989.650422116736
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 7031.883083087312
 - Absolute error: 127.19685275232314
 - Relative error: 0.01842181505562086



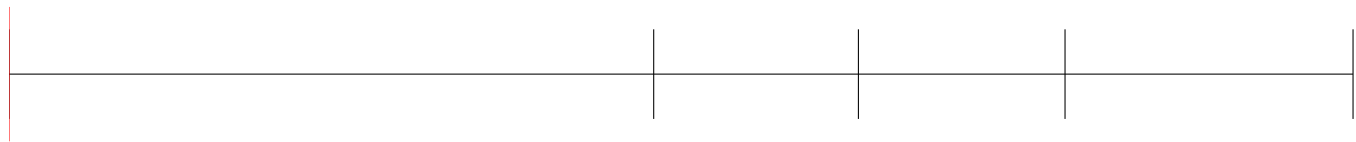
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



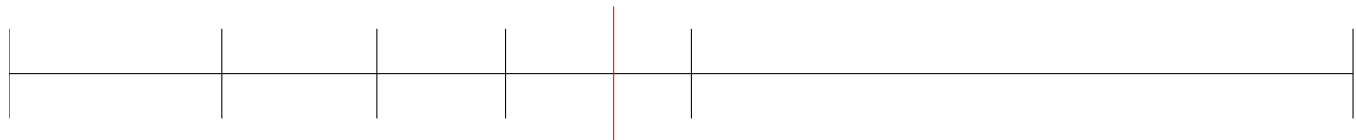
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



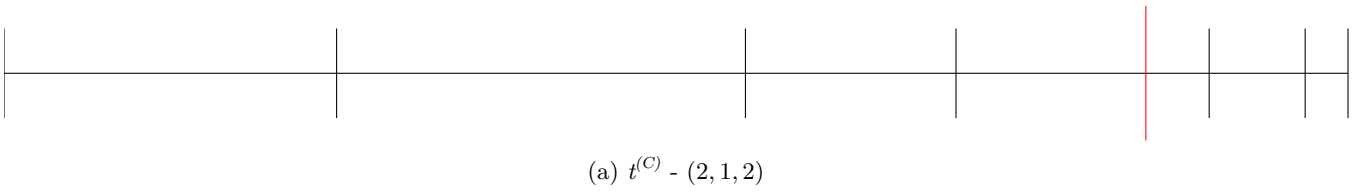
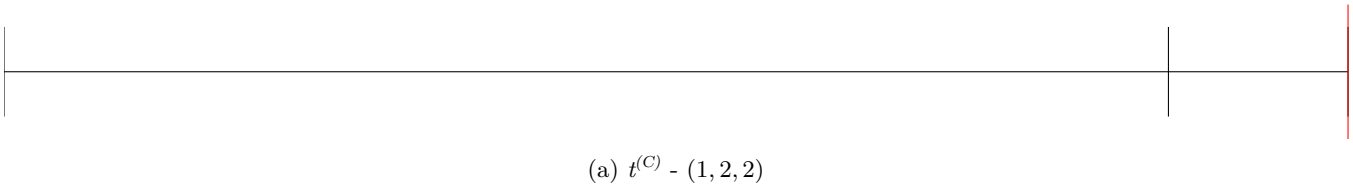
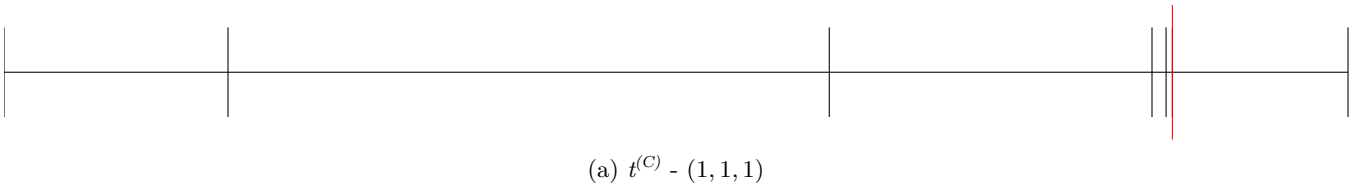
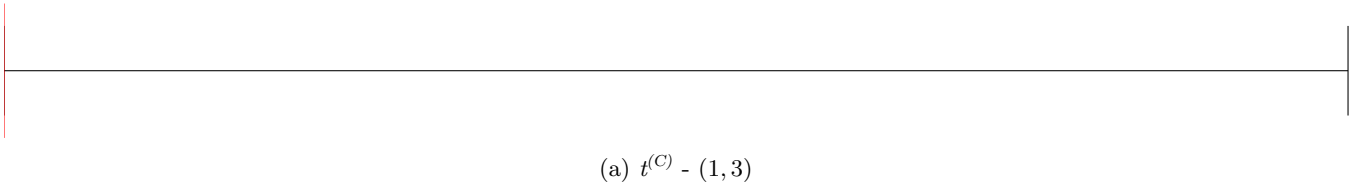
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

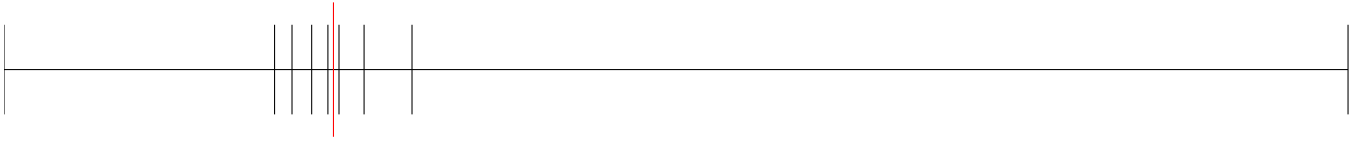
- $(1, 1, 1)$
 - Value: 70.301818
 - Error: 1.107557797540196
 - Relative error: 0.015509977388134
- $(1, 2, 2)$
 - Value: 69.345028
 - Error: 0.000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 131.697605
 - Error: 4.313718140889875
 - Relative error: 0.031715875227973

5.2 A_{CU}

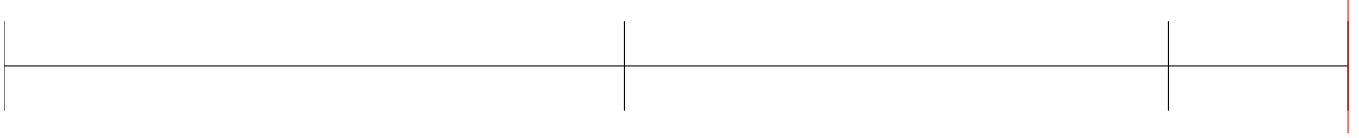
- $cu, 1$
 - Value: 3.180942
 - Error: 1.613847925716230
 - Relative error: 0.336583659708964
- $cu, 2$
 - Value: 37.478751
 - Error: 0.383042620517479
 - Relative error: 0.010116864075006

5.3 A_{HU}

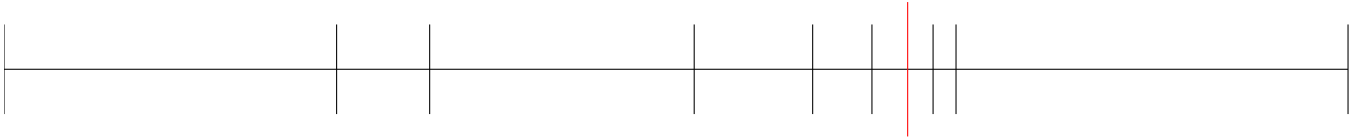
- $hu, 1$
 - Value: 13.227179
 - Error: 0.092706320672210
 - Relative error: 0.006959993799339



(a) $A^\beta - (1, 1, 1)$



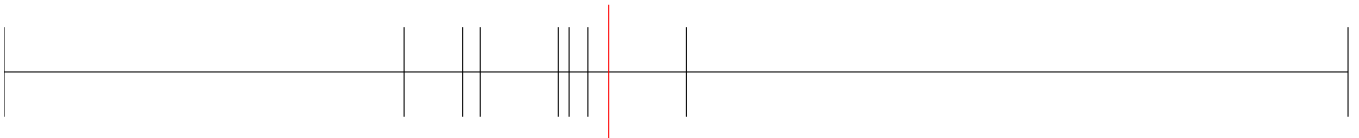
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

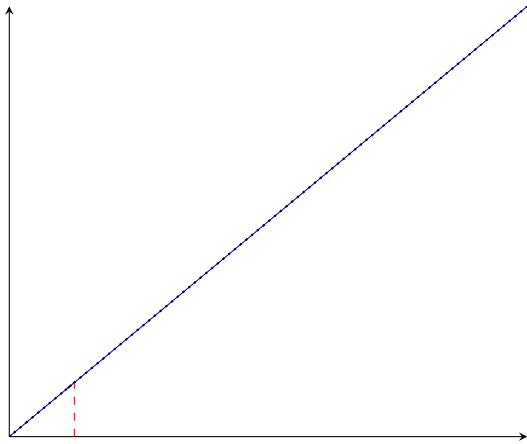


(a) $A^\beta - hu, 1$

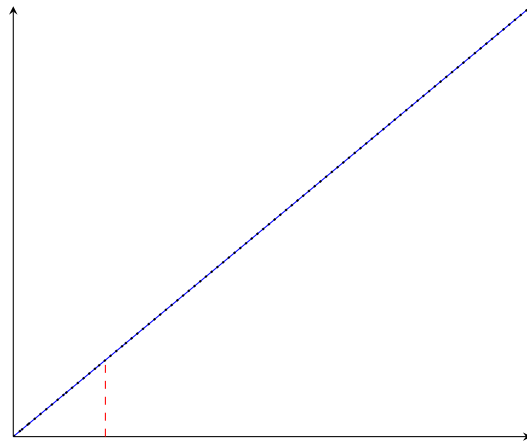
6 A^β Breakpoints

- (1, 1, 1)
 - q : 70.301818
 - $\hat{q}^\beta$: 70.301818
 - q^β : 70.301818
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 69.14107665413653, 69.89902342903967, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.345028
 - $\hat{q}^\beta$: 69.345028
 - q^β : 69.345028
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 131.697605
 - $\hat{q}^\beta$: 131.697605
 - q^β : 131.697605
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 147.48374761404892, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 3.180942
 - $\hat{q}^\beta$: 3.180942
 - q^β : 3.180942
 - Breakpoints: [0, 2.9819181007907165, 4.432280503220532, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.478751
 - $\hat{q}^\beta$: 37.478751
 - q^β : 37.478751

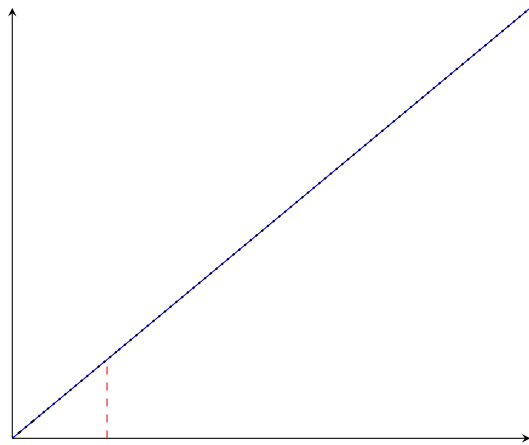
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.227179
 - $\hat{q}^\beta$: 13.227179
 - q^β : 13.227179
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 13.192252946941991, 13.244985069124171, 13.30871858348944, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



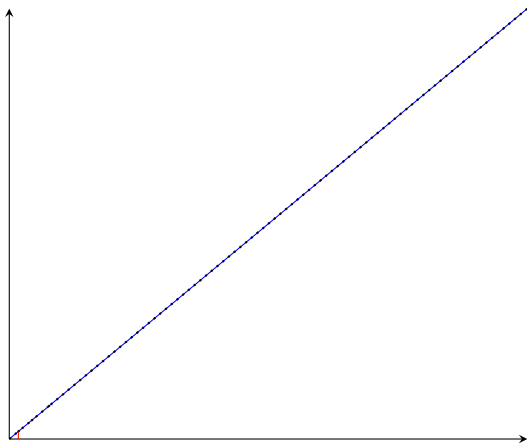
(a) $q^\beta - (1, 1, 1)$



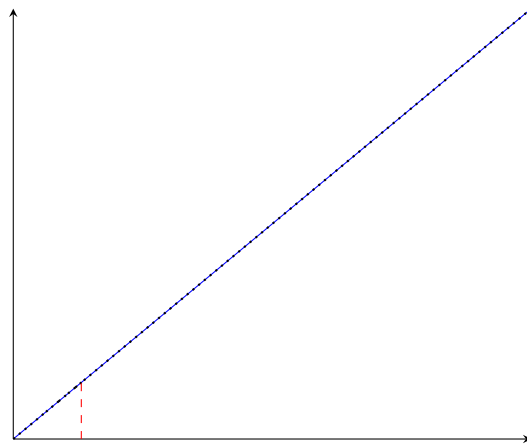
(b) $q^\beta - (1, 2, 2)$



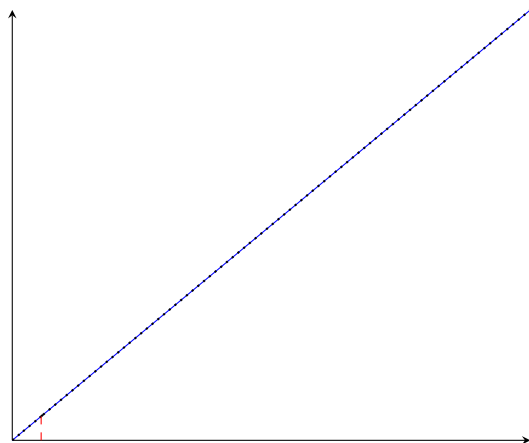
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1538.5911064906686
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008088138481334745	$b_{2,1,1}^{C,in}$	267.17430774259145
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.004340277777777778	$b_{1,1,2}^{C,out}$	1538.5911064906686
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	267.17430774259145
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	557.1548250268942		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	466.16891151778816		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	417.2070597455825		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4906.686245775449		
$f_{1,2,1}^H$	7.548748070423766	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	3.752661235343094	$b_{1,2,1}^{H,out}$	4906.686245775449		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.4685536278409863	$b_{2,2,1}^{H,out}$	0.0		